



PERTH MODERN SCHOOL
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Test 2

Proportion, Functions, Relations & Transformations

Semester One 2019

Year 11 Mathematics Methods

Calculator Assumed

Name:	
Teacher:	

Date: Friday 12th April 7.45am

You may have a formula sheet and 1 page (1 side) of notes for this test.

Total_____ / 41

Total Marks:41

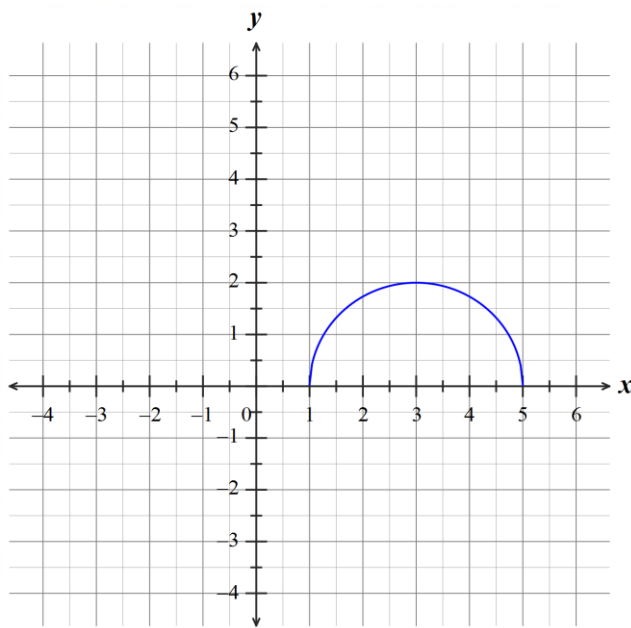
Time: 45 Minutes

Question 1**(3 marks)**

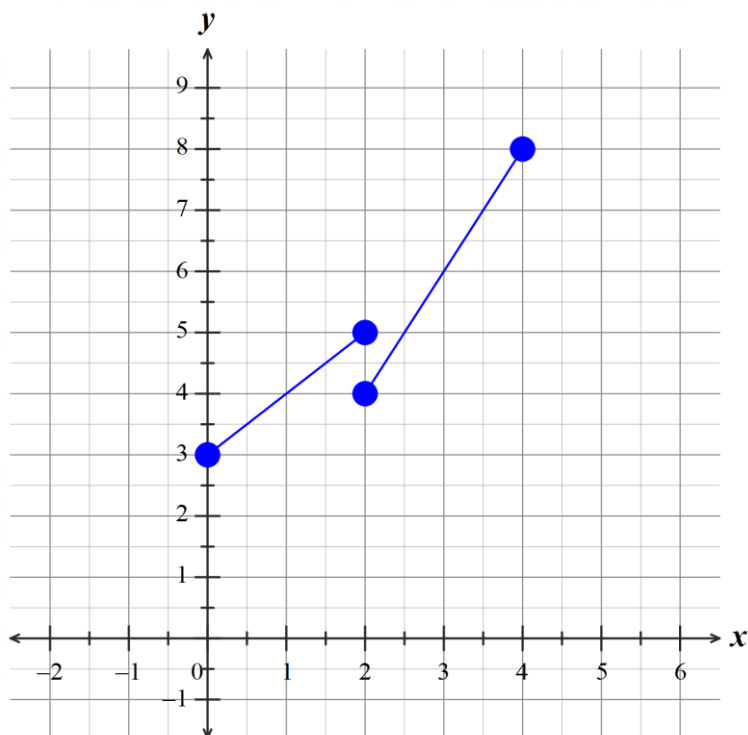
State whether the following relations are functions.

a) $\{(0, 0), (1, 1), (1, -1), (4, 2), (9, 3)\}$

b)



c)



Question 2**(4 marks)**

Given that y is directly proportional to the square of x . When $y = 12$, $x = 4$, find

a) the constant of variation

(2 marks)

b) the value(s) of x when $y = 27$

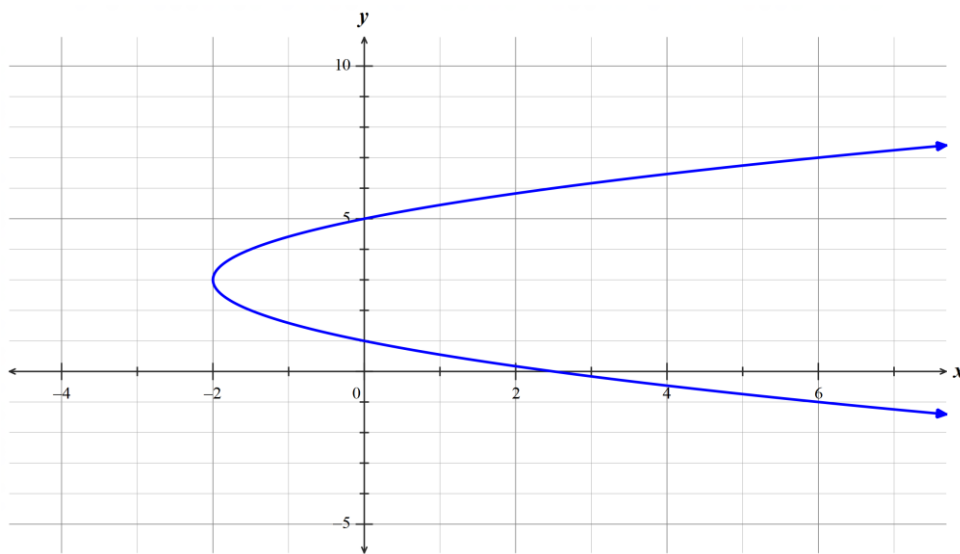
(2 marks)

Question 3**(8 marks)**

- (a) Find the radius and the coordinate of the centre of the circle with equation $x^2 + y^2 - 4x - 6y - 3 = 0$. Show your working.

(3 marks)

- (b) The variables x and y are related as demonstrated by this graph.



- i) Determine the equation of the graph above. **(3 marks)**

- ii) State the domain. **(1 mark)**

- c) From **(a)** and **(b)**, what features of their graphs clearly indicate that x is not a function of y ? **(1 mark)**

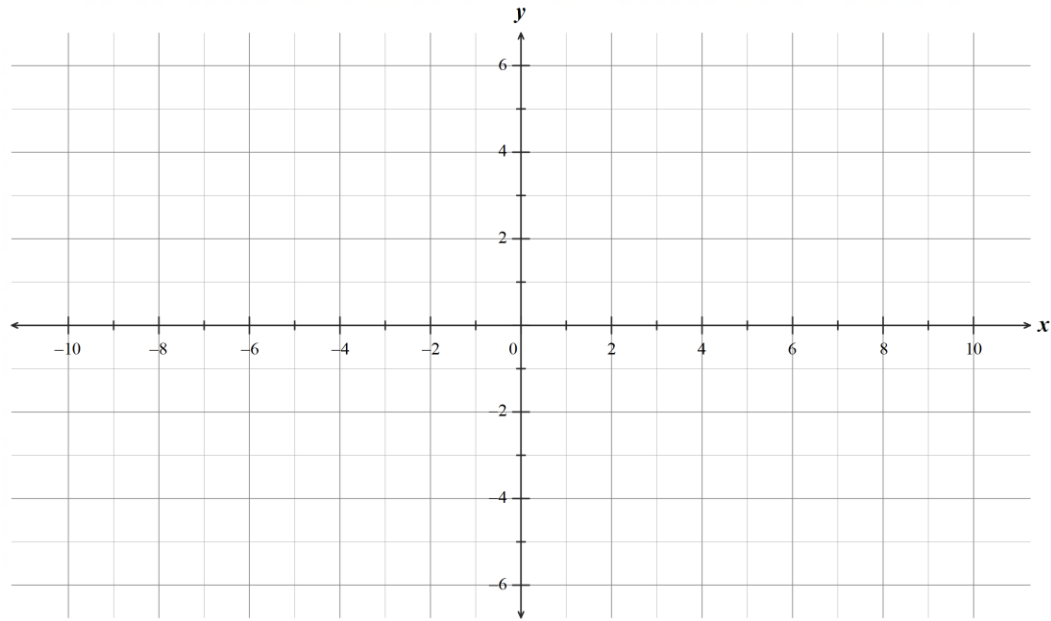
Question 4**(6 marks)**

The function $f(x) = \sqrt{x}$ is transformed into $g(x) = k\sqrt{ax + b} + c$ by the following sequence of transformations.

(a) Sketch the following transformation of $f(x)$.

'A translation 5 units in the positive x -axis followed by a translation of 2 units in the positive y -axis.'

(2 marks)



(b) Determine the equations of the resulting function $g(x)$.

i) A translation 3 units in the direction of the negative y -axis followed by a reflection about the x -axis. (2 marks)

ii) A dilation parallel to the positive x -axis of factor 2 followed by a translation 4 units in the direction of the positive x -axis (2 marks)

Question 5**(9 marks)**

Consider the functions f and g where $f(x) = ax^2 + bx + c$ and $g(x) = f(2x + 3)$.

a) Given $f(-2) = 0$, $f(5) = 0$ and $f(2) = 3$, determine the rule for $f(x)$. (3 marks)

b) Express the rule for $g(x)$ as a polynomial. (3 marks)

c) The coordinate $(1, 3)$ lies on $f(x)$. Determine the coordinate for $f(x) - 4$. (1 mark)

d) Describe the sequence of transformations that would transform $f(x)$ to $g(x)$. (2 marks)

Question 6**(4 marks)**

The time (t) in hours required to construct a retaining wall varies inversely to the number of workers (w) being employed. An engineer estimates that it will take 8 workers 180 hours to construct a retaining wall. [Assume that all workers work at the same rate.]

- a) If the retaining wall must be constructed in 150 hours, how many extra workers will need to be employed? (3 marks)

- b) If only 6 workers are available, how long will they take to construct this wall? (1 mark)

Question 7**(7 marks)**

(a) Express $f(x) = \frac{6x-15}{x-3}$ into the form $f(x) = \frac{a}{x-h} + k$.

(2 marks)

(b) Determine the coordinate of the x -intercept.

(1 mark)

(c) State the asymptotes of $f(x)$.

(2 marks)

(d) Hence, determine the range of $f(x)$.

(2 marks)